

National University of Computer and Emerging Sciences, Lahore Campus

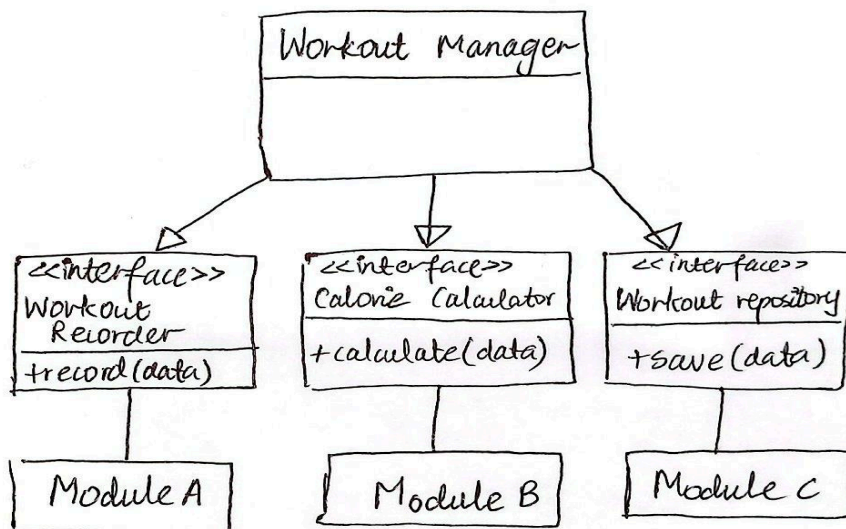
	Course Name	Software Design & Analysis	Course Code	CS3002
	Name	solution	Exam Type	Quiz 4.2
	Roll.No	22I-1234	Total Marks	20
	Section:	5J	Weight	tba

Q1(a):

Consider a mobile fitness application that tracks user activities. One class in the application is WorkoutManager. This class performs three unrelated responsibilities: It records workout data (like steps, distance), calculates calories burned using a formula and it saves the workout history to the database. Several modules of the app use these features. Module A only needs the workout recording feature. Module B only needs the calorie calculation feature. Module C only needs the saving-to-database functionality. Refactor (improve) the design in the light of SOLID principles and draw the solution diagram. Also mention the SOLID principle violated in the original design.

Isolated?

SRP & ISP violated



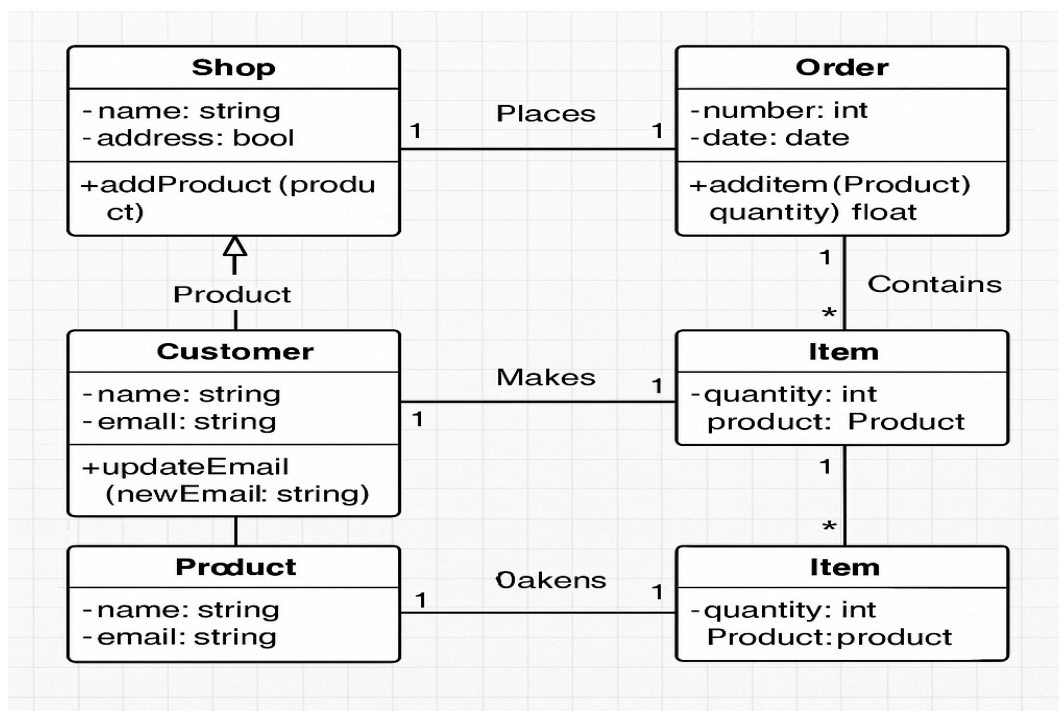
(b) A class `ReportGenerator` creates reports, formats them, and also sends them via email. Which SOLID principle is being violated?

- A. Liskov Substitution Principle
- B. Open-Closed Principle
- C. Single Responsibility Principle
- D. Dependency Inversion Principle

(c) A `NotificationService` directly creates objects of `EmailSender`, `SMSSender`, and `PushSender` inside itself. Whenever a new notification type is added, the class must be modified. Which SOLID principle is violated?

- A. Open-Closed Principle
- B. Dependency Inversion Principle
- C. Liskov Substitution Principle
- D. Interface Segregation Principle

Q2: Without making any assumptions , use the assumptions provided in the design class diagram above to determine the values of the OO metrics for the classes specified in the diagram below :



Sr#	Class	Metrices	Value
1	Shop	Weighted Method per Class	1
2	Order	Weighted Methods per Class	1
3	Customer	Weighted Methods per Class	1
4	Product	Depth Of Inheritance(DIT)	0
5	Item	Depth Of Inheritance(DIT)	0
6	Shop	Depth Of Inheritance(DIT)	0
7	Shop	Number Of Children (NOC)	1
8	Order	Number Of Children (NOC)	0
9	Item	Coupling between objects (CBO)	3
10	Product	Coupling between objects (CBO)	2